🔑 **Answer Key — Teacher Copy · Lesson 6-3**

MSTAR PRACTICE WORKSHEET

Lesson 6-3: Explore Numerical Expressions with Exponents

Grade 6 Mathematics · Standard 6.AT.5

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: D

What is the value of 4^3?

(A) 7

(B) 12

(C) 43

(D) 64

$4^3 = 4 \times 4 \times 4 = 64$.

Question 2 SELECTED RESPONSE — CORRECT: C

What is the value of 6^2?

(A) 8

(B) 12

(C) 36

(D) 62

$6^2 = 6 \times 6 = 36$.

Question 3 SELECTED RESPONSE — CORRECT: B

Which expression shows 5^3 as repeated multiplication?

A. $3 \times 3 \times 3 \times 3 \times 3$

B. $5 \times 5 \times 5$

C. 5×3

D. $5 + 5 + 5$

5^3 means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.

Question 4 SELECTED RESPONSE — CORRECT: C

Evaluate: $3 \times (4 + 2)^2$

A. 36

B. 54

C. 108

D. 144

$4 + 2 = 6$. $6^2 = 36$. $3 \times 36 = 108$.

Question 5 SELECTED RESPONSE — CORRECT: D

The drum loop plays 3^4 beats. What is the value of 3^4?

A. 7

B. 12

C. 27

D. 81

3^4 means use 3 as a factor 4 times: $3 \times 3 \times 3 \times 3 = 81$. It is NOT $3 \times 4 = 12$.

Question 6 SELECTED RESPONSE — CORRECT: B

A synth pattern repeats 7 × 7 × 7. How do you write that as a power?

A 3^7

B 7^3

C 7×3

D 21^3

The base 7 is used as a factor 3 times, so the power is 7^3. $7^3 = 7 \times 7 \times 7 = 343$.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

Which value is equal to the power 3^4?

A 7

B 12

C 64

D 81

The base is 3 and the exponent is 4, so 3 is used as a factor four times: 3 × 3 × 3 × 3 = 81.

- **A:** That adds 3 and 4. The exponent tells you to multiply 3 by itself four times.
- **B:** That multiplies the base by the exponent. An exponent counts how many times 3 is used as a factor.
- **C:** That is four used three times, with base and exponent swapped. Here the base 3 is used four times.

Part B — correct answer: C

Which reasoning shows why your answer to Part A is correct?

A Multiply the base by the exponent: 3 × 4 = 12.

B Add the base and the exponent: 3 + 4 = 7.

C The exponent 4 tells how many times to use the base 3 as a factor: 3 × 3 × 3 × 3 = 81.

D Use the exponent as the base and the base as the exponent: 4 × 4 × 4 = 64.

An exponent counts repeated factors of the base. It does not tell you to multiply, swap, or add the two numbers.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about the power 2^5.

- **A** The base is 2.
- **B** The exponent is 5.
- **C** It means 2 × 2 × 2 × 2 × 2.
- **D** Its value is 10.
- **E** Its value is 32.
- **F** It means 5 × 5.

In 2^5 the base 2 is used as a factor 5 times: 2 × 2 × 2 × 2 × 2 = 32. A value of 10 comes from multiplying 2 × 5, and 5 × 5 swaps the base and the exponent.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

Model answer: Doubling every hour means multiplying by 2 once per hour, so after 6 hours the count is 2^6 = 2 × 2 × 2 × 2 × 2 × 2 = 64 bacteria. Adding 2 each hour instead would give only 1 + 12 = 13. Repeated multiplication compounds — each hour multiplies everything already there — while repeated addition only ever adds the same fixed amount.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.